

Synthesizing Strategic Insight: Evaluating Fine-Tuning and Agentic RAG for Professional-Grade Sports Commentary

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1 Introduction

Formula one is a sport deeply grounded in research and development, that generates billions in revenue each year. With almost a billion international fans watching each race, it is a shame that the sport hasn't become more accessible. In addition, F1 teams are progressively becoming more data-driven and reliant on their strategy and development of their teams and cars. This paper proposes an approach that drives strategic insight via commentary suitable for the consumer and for the teams aiming to acquire assistance with data interpretation. From this paper, consumers of Formula 1 with consumer level hardware should be able to access and use our solution, and teams may be able to find unseen insight promoting improved strategy.

Previous solutions and or proposed technology have not existed for this work, and most likely is soon to be considered with many of the Formula teams creating strategic partnerships with world leaders in artificial intelligence. This paper aims to provide direction towards a valuable asset that may become standard within not only Formula 1 teams' strategy, however additionally adds to the existing corpus of research regarding autonomous sports commentary generation by incentivizing synthesized insight. We believe that via a learned intelligent mapping from live telemetry and previous commentary, a fast consumer grade tool can provide useful intuition.

This paper illustrates the next step in Formula 1 commentary, and a different approach to other sports autonomous sports commentary generation. This is done with the model being built for synthesized insight, in lieu of providing a mere verbatim narration of the visual elements. Furthermore, this approach is optimized to perform on consumer level hardware, ensuring accessibility for consumers, and less powerful potentially embedded applications.

2 Related Work

Automatic Sports Commentary Generation

- [1]
- [2]
- [3]

The research regarding intelligent autonomous sports commentary is extremely sparse, as often regarded within each source’s related works. Formula one is not short of research; however, within this domain, the work is within infancy. Our work, which involves applying such commentary to Formula 1, is a challenge only made possible via similar works done within baseball and basketball. The work undertaken by Lee on [2] provides an insight on how we can make our commentary more insightful from an automated narrative.

DPO vs. RLHF

- [4]
- [5]
- [6]

A prime motivation within commentary been to provide listeners with insightful, most importantly entertaining dialogue. In previous contexts regarding large language models and the broader scope of natural language tasks, we have seen advancements in tailoring responses to match human preferences. A breakthrough within this realm has come from the explorations undergone by Ouyang [5]. Although the benefits from introducing RLHF can greatly improve the commentary generated by an approach found by Rafailov [4] would reap the benefits of such implementation without requiring as much overhead. Both are notable potential additions upon our proposed methodology; however, with the added cost of compute or dataset additions, we are unable to implement such without risk of straying from the affordance of being consumer feasible and achievable within a month of our development. With more time, we would aim to implement DPO into our system.

Proper Dataset Creation

- [7]

In contrast to existing datasets and achievements from similar works such as Zhang [7], Formula 1 constantly makes regulation overhauls on average every 6-8 years. This creates an issue in dataset creation, as we are unable to use a significantly large set of previous years’ telemetry data. Additionally, we were unable to ethically source visual data, which resulted in an absence of visual media within our dataset. In the creation of our dataset, we used strategies learned from Zhang to create a suitable dataset based upon the most up to date regulations within our targeted sport. We relied on historical telemetry data from OpenF1 and publicly available ground truth commentary.

3 Methodology

Methods - describe the implementation of your proposed idea (e.g., description of your model, system, or how survey papers will be collected) so that:

- A reader could reproduce your set-up
- A reader could understand why you made your design decisions

4 Experiments

4.1 Latency and User Feasability

As our solution is required to run simultaneously with a live race, latency is exceptionally important. What is not fair about this undergoing analysis on this metric is that the hardware in which our system executes can quite obviously boost our performance and lower our latency; this is why we will be aiming for common consumer hardware for testing. The system will run on three different hardware setups, modeling what we believe to be the most common consumer hardware as of 2026. As the tech world has been largely buying mac mini's to run AI solutions, we will be undergoing testing on a 2022 MacBook Pro M1 chip with 16 GB of unified memory, a M4 MacBook Air with 16 GB of unified memory, and additionally a Linux laptop with an Nvidia RTX 4060 to gauge higher-level hardware. We will be aiming for a latency from cognition (when the telemetry packet has been recieved) to final output (when the model has began generation) of less than 1.5 seconds across all devices.

4.2 Our Solution vs. Frontier Models

Additionally, we will be conducting A/B testing between our solution and Frontier models, including Google Gemini 3 Fast, Claude Sonnet 4.6, and ChatGPT 5.4. We will be using Evaluation scores from BERTscore, ground truth accuracy for hallucination analysis, and a human survey-based preference for subjective quality. The goal of this evaluation is to capture quantitative data of our open source solution against frontier alternatives. With this, we will be able to see how we can tune our solution to improve matching the expectations of consumers, and possibly see if it makes more sense to use frontier models within our system.

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